

EDF R&D	SYRTHES 5 Tutorial	Version 1
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Chapter 1

Getting started with conduction in a square and a disk

Conduction in dimension 2

Square and disk

1.1 What is the problem ?

We would like to compute the temperature in a component formed by a square intersecting a disk.

1.1.1 Geometrical description

The solid domain consists in a piece formed by 2 component : a square and a disk : square and disk are made of steel but intersection between them has the charactics of an insulator.

Dimensions of the square is around 1 meter.

Geometrical characteristics are shown on figure 7.1

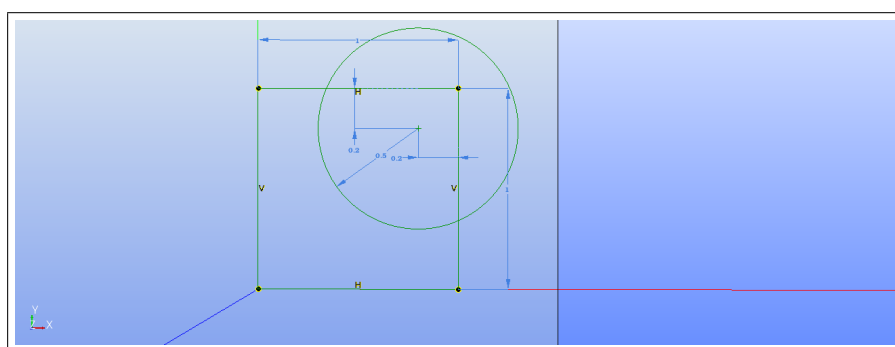


FIGURE 1.1 – Sketch of the problem

1.1.2 Physical description

Density and specific heat are set to :

- Steel for the square and the disk : $\rho = 7700 \text{ kg/m}^3$ $C_p = 460 \text{ J/kg}^\circ\text{C}$ $k = 25 \text{ W/m}^\circ\text{C}$
- insulator for the intersection between square and disk : $\rho = 1000 \text{ kg/m}^3$ $C_p = 1000 \text{ J/kg}^\circ\text{C}$ $k = 1 \text{ W/m}^\circ\text{C}$

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1.1.3 Initial conditions and boundary conditions

The initial temperature is $20^{\circ}C$. Boundary conditions are :

- square boundary : $T = 50^{\circ}C$, $h = 1000 \text{ W/m}^2/^{\circ}C$
- disk boundary : $T = 10^{\circ}C$, $h = 1000 \text{ W/m}^2/^{\circ}C$
- other boundaries : $\text{flux} = 0$.

1.2 How to do that ?

1.2.1 How to organize the study

This section is facultative, we suggest how to manage files and directories of your study. This is only advice, advanced users can do as they want...

- create a new directory for your study : `mkdir conduction-square2D`
- go inside : `cd conduction-square2D`
- create a new directory for the creation of the mesh : `mkdir salome`

1.2.2 Creating mesh

If the geometry stays very simple and do not create any trouble, attention must however be paid to the references allowing to identify materials and boundary conditions. We used SALOME to define the geometry and create the mesh.

In the directory `conduction_square2D/salome`, run SALOME. You are ready to create your mesh. Save your SALOME-study and export your mesh to MED format in this directory.

In order to define the different boundary conditions and material properties, groups of faces and edges have been created.

You can create your own mesh, but below, are described the characteristics of the mesh provided in the SYRTHES distribution :

`/.../syrthes5.x/arch/your_arch/share/syrthes/tests/1-conduction-square2D.`

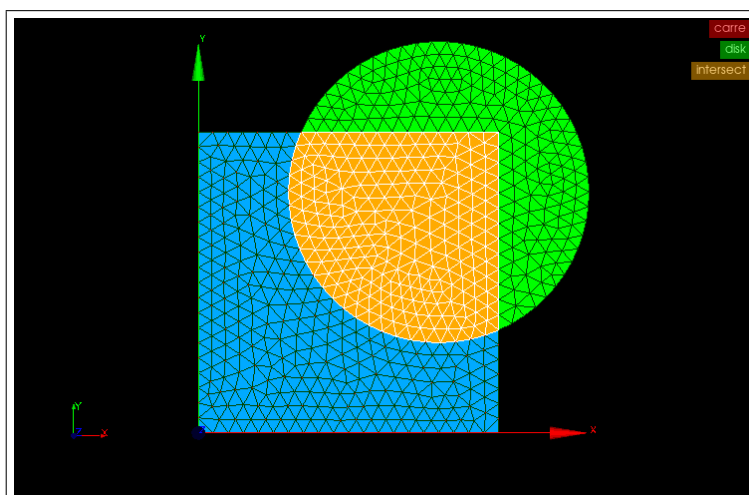


FIGURE 1.2 – Mesh

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1.2.3 Create your SYRTHES-study

Go back to the initial directory : `conduction-square2D`
 If not already done : source the SYRTHES environment (Linux only) :
`source ../../syrthes5.0/arch/myarch/bin/syrthes.profile`
 Run the SYRTHES-gui : `syrthes.gui`



FIGURE 1.3 – SYRTHES Managing your cases

Create a new case : `syrthes`

Now, all your calculation will be managed by the SYRTHES Graphic User Interface.

1.2.4 Main view

Give a title to your study. The dimension of the problem is 3D.

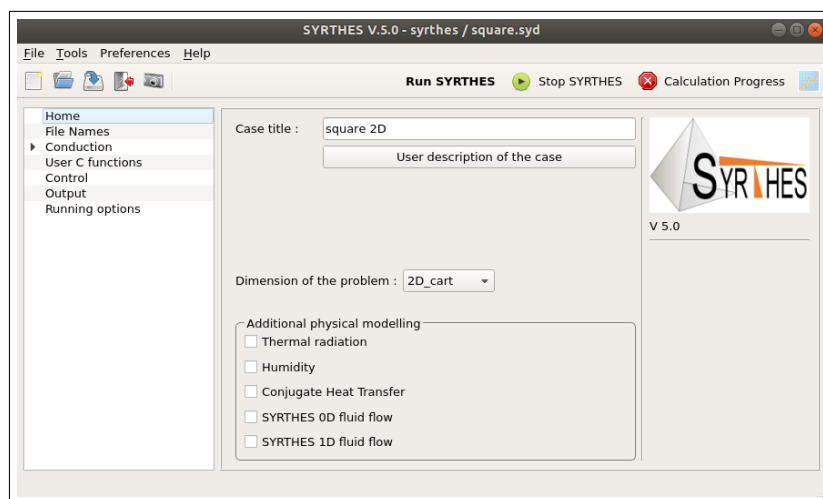


FIGURE 1.4 – SYRTHES Main View

Save your data file :

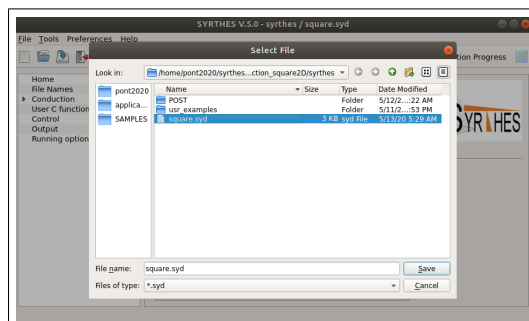


FIGURE 1.5 – SYRTHES Save your data file

1.2.5 File Names

- Click on the next item in the menu on the left : File Names
- Select your conduction mesh : **conduction-square2D/salome/square.med**

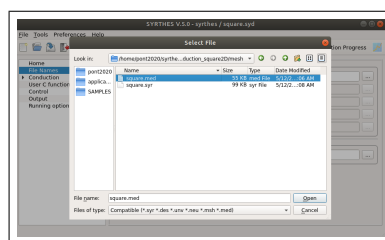


FIGURE 1.6 – Select a mesh

A conversion of the file format is done automatically and you should get the message :

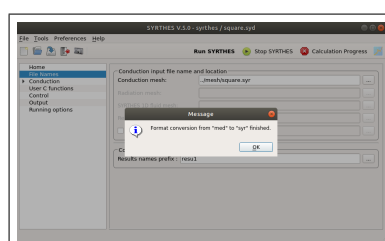


FIGURE 1.7 – SYRTHES File format conversion OK

Finally, you can choose the name of your results files (a name without extension ; SYRTHES will create different files with the same radical but different extensions depending on the type of files). By default the name is `resu1`.

The file Names looks as shown below :

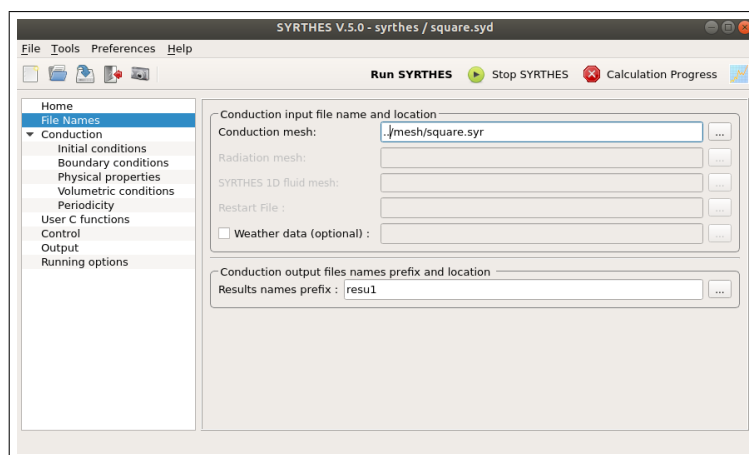


FIGURE 1.8 – SYRTHES File Names window

1.2.6 Input data for conduction computation

1.2.6.1 Group names and references numbers

While SALOME is using group names to distinguish different parts in the mesh, SYRTHES is using reference numbers. Group names and reference numbers are included in the mesh file `square.med`. The links between both are given in an additional file (`.syr_descr`) created while converting the MED file to the SYRTHES file. You can open this ASCII in the SYRTHES GUI : menu “Tools” and “Open Desc”.

WARNING : depending on SALOME version, pairs (group-name , number) could change. So make sure you have a look on the description file and adapt numbers before proceeding.

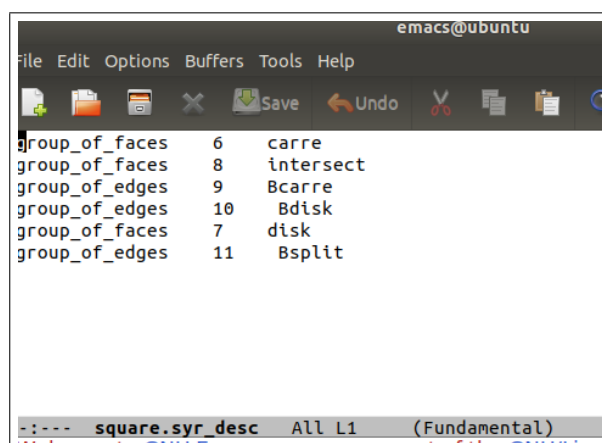


FIGURE 1.9 – Mesh description file (salome/square.syr_descr)

1.2.6.2 Initial conditions

Unroll the conduction menu by clicking on the arrow, and select the first item : Initial conditions

Set the initial temperature (20°C), and the list of volumes considered. If all the volumes are affected by the same initial condition, you can put “-1” instead of an explicit list. You can add a comment in the last column (optional).

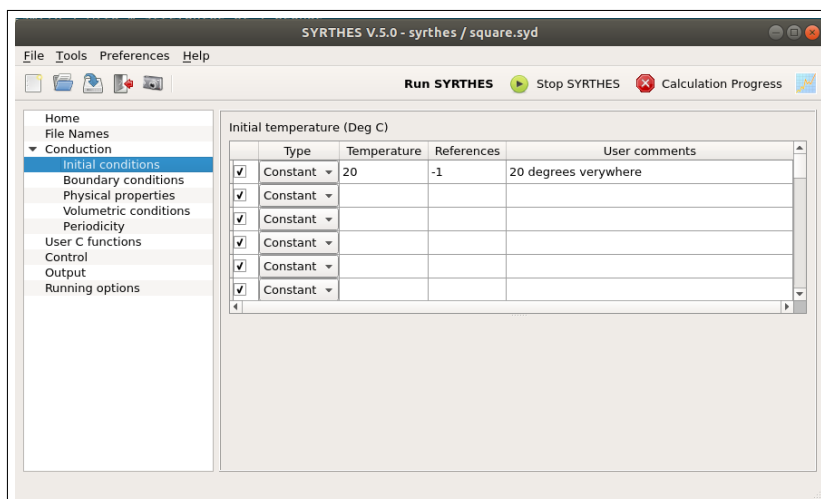


FIGURE 1.10 – SYRTHES - Initial conditions

1.2.6.3 Boundary conditions

We now want to set an heat exchange coefficient along the square and disk borders. T

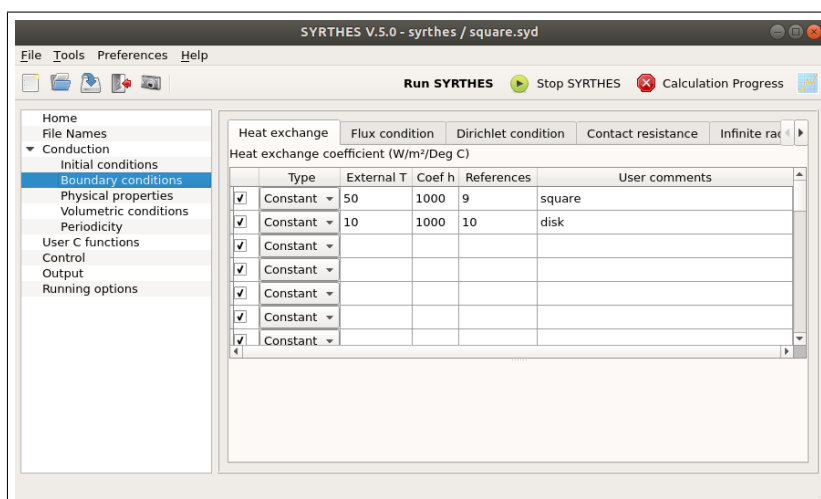


FIGURE 1.11 – SYRTHES - Boundary conditions

1.2.6.4 Physical properties

The model has different material properties.

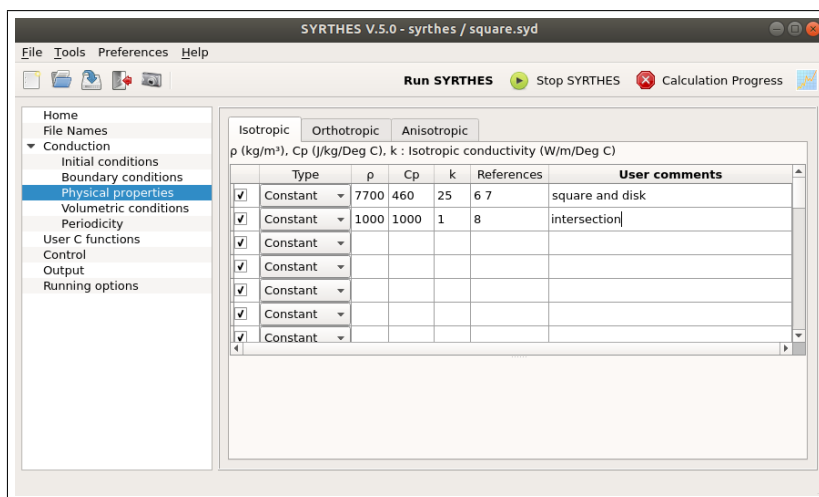


FIGURE 1.12 – SYRTHES - Isotropic conductivity

Now all physical parameters are defined, for this simple case you can jump directly to the Control Window.

1.2.7 Control

We have to set the time step. The stationnery state is computed as the converged state of an transient computation. The time step has to stay reasonable, if a fair precision is required during the transient, indeed the time error is more or less proportionnal to the time step retained. In the present case, and considering the mesh, a time step from around 300 s seems reasonable.

We compute 300 time steps.

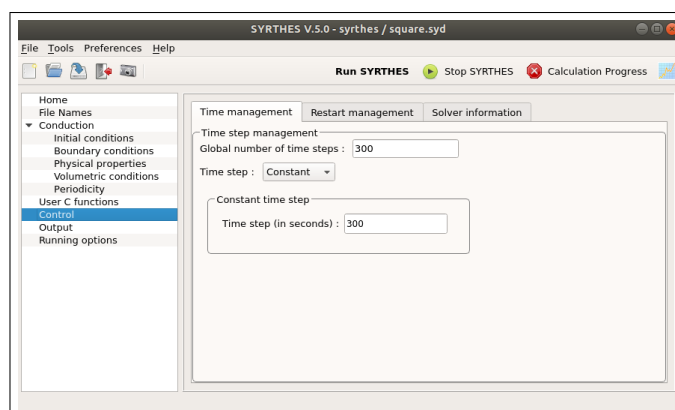


FIGURE 1.13 – SYRTHES - Control window

Generally, there is no need to change the default values provided for the solver.

1.2.8 Output

In this section, we are going to define the type of results to be generated by SYRTHES. Whatever the options you will have a result file with the temperature on all nodes of the mesh.

But, sometimes, it is also advisable to define some thermal probes : during the run, you will be able to follow the temperature evolution at some strategic points of your domain. Set the coordinates of points in your domain. Here we have selected 2 points. Then, define the frequency you want to save the values of the probes on file : here, 1 means “ every time step”.

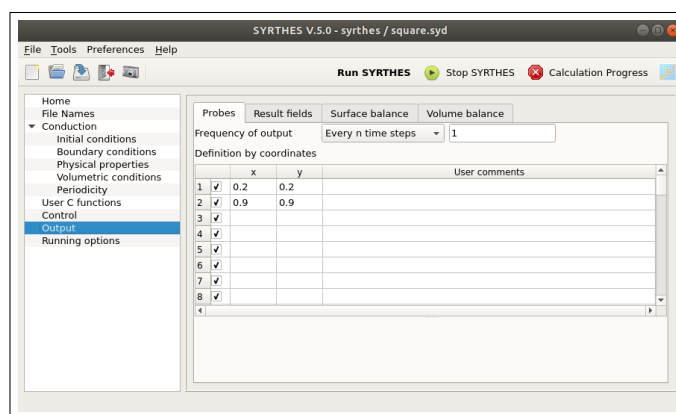


FIGURE 1.14 – SYRTHES - Defining thermal probes

1.2.9 Running options

This is the last step : give a name for the listing file. SYRTHES will give you some information about the calculation (option summary, solver convergence,...) Furthermore, according the kind of computer at your disposal and how big is the case, you can try a parallel computation, setting the number of processors to 3 or 4 (but here, as the element number of this mesh is very low, the benefit in term of CPU time will be very low).

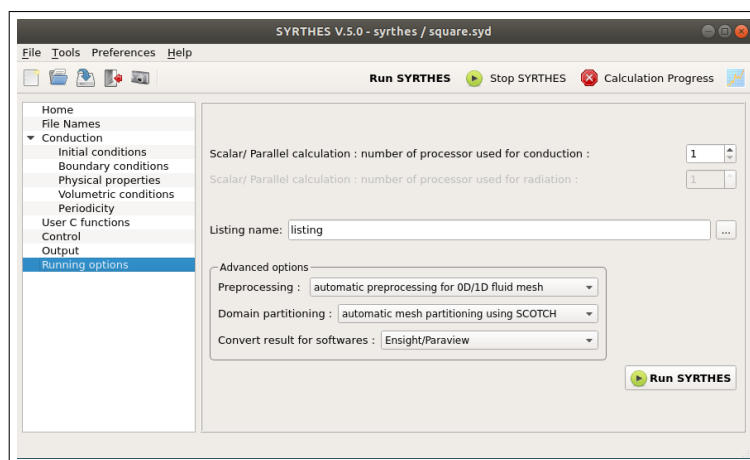


FIGURE 1.15 – SYRTHES - Running options window

1.2.10 You're ready to run SYRTHES !

Click "Run SYRTHES", the "calculation progress" window appears and you can display the evolution of the temperature at the 3 points defined previously.

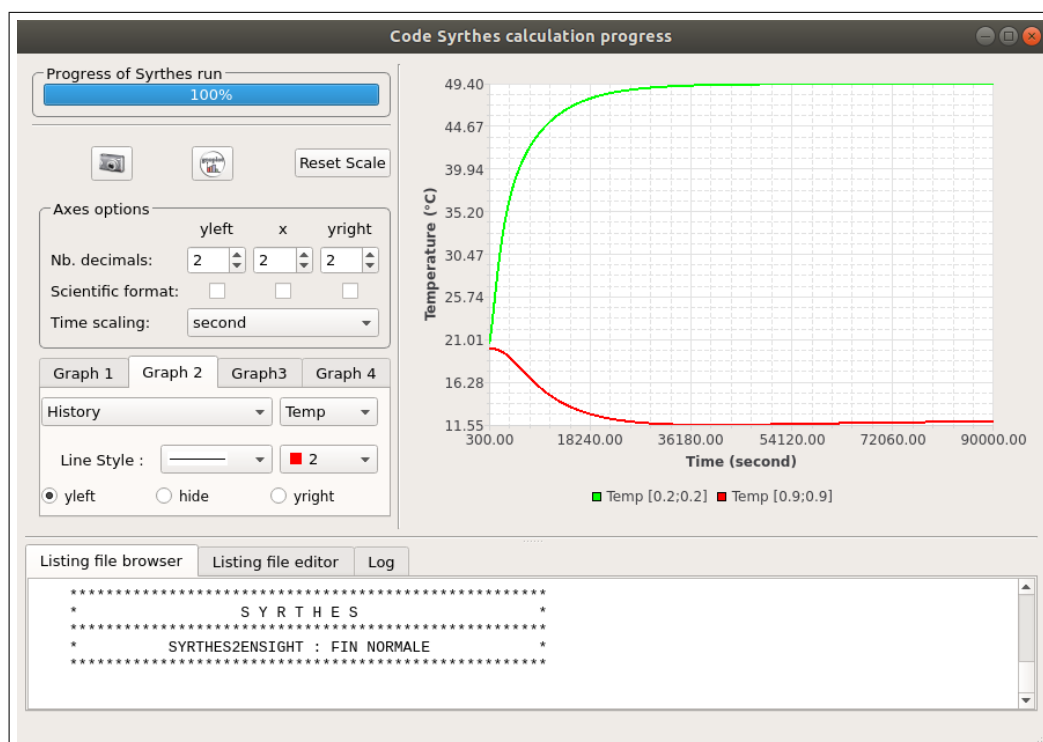


FIGURE 1.16 – SYRTHES Calculation progress window

The 3 tabs display :

- Listing file browser : displays the 200 last lines of the listings
- Listing file editor : whole SYRTHES listing file
- Log : system messages (usefull in case of error during user files compilation)